

ATNA

Air Traffic Network Analysis

Interface walkthrough

Structural criticality and scenario modelling for the U.S. domestic flight network. Every panel is a live capture of the running application.

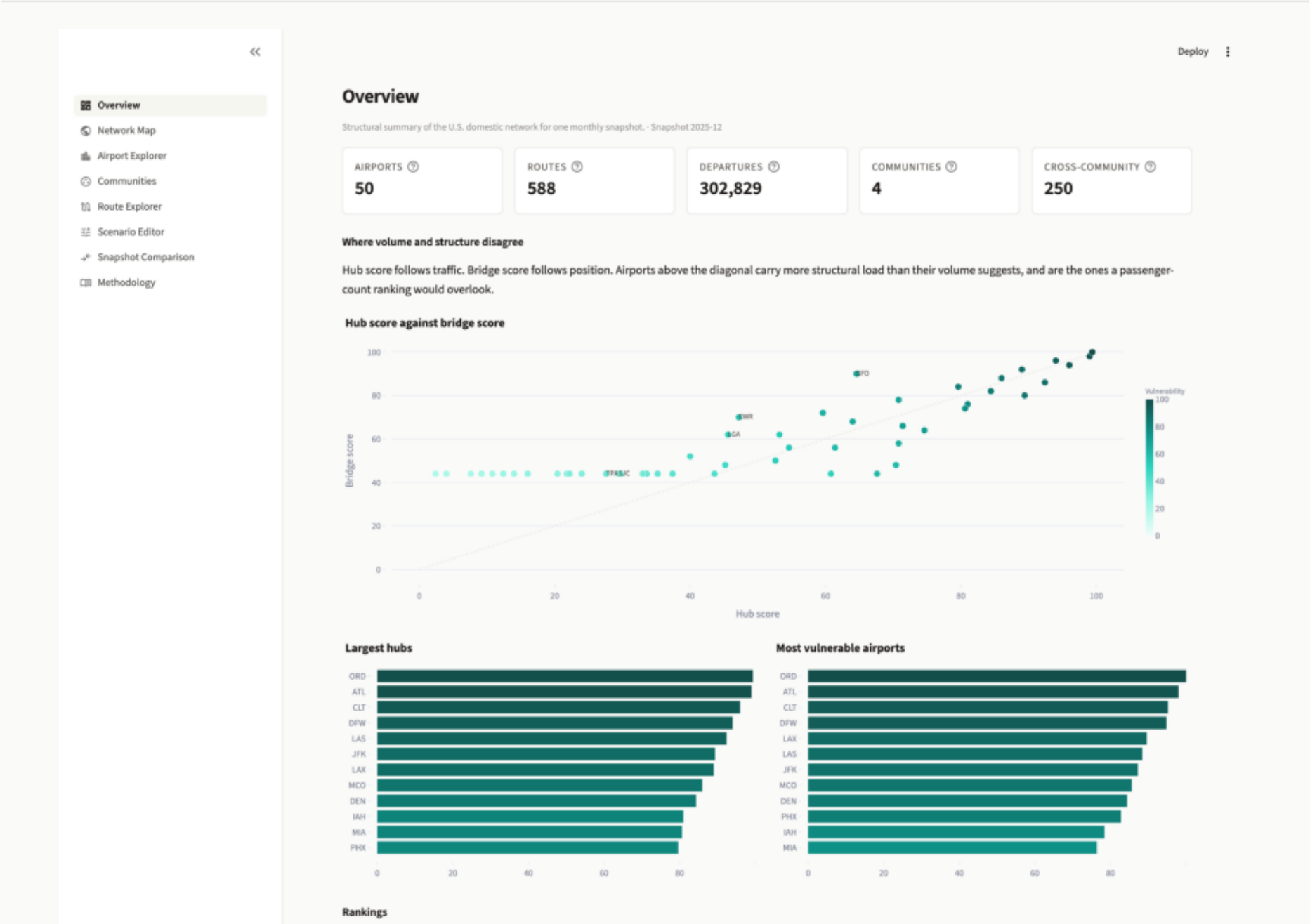
Snapshot 2025-12

Pages 9

Generated 2026-08-27

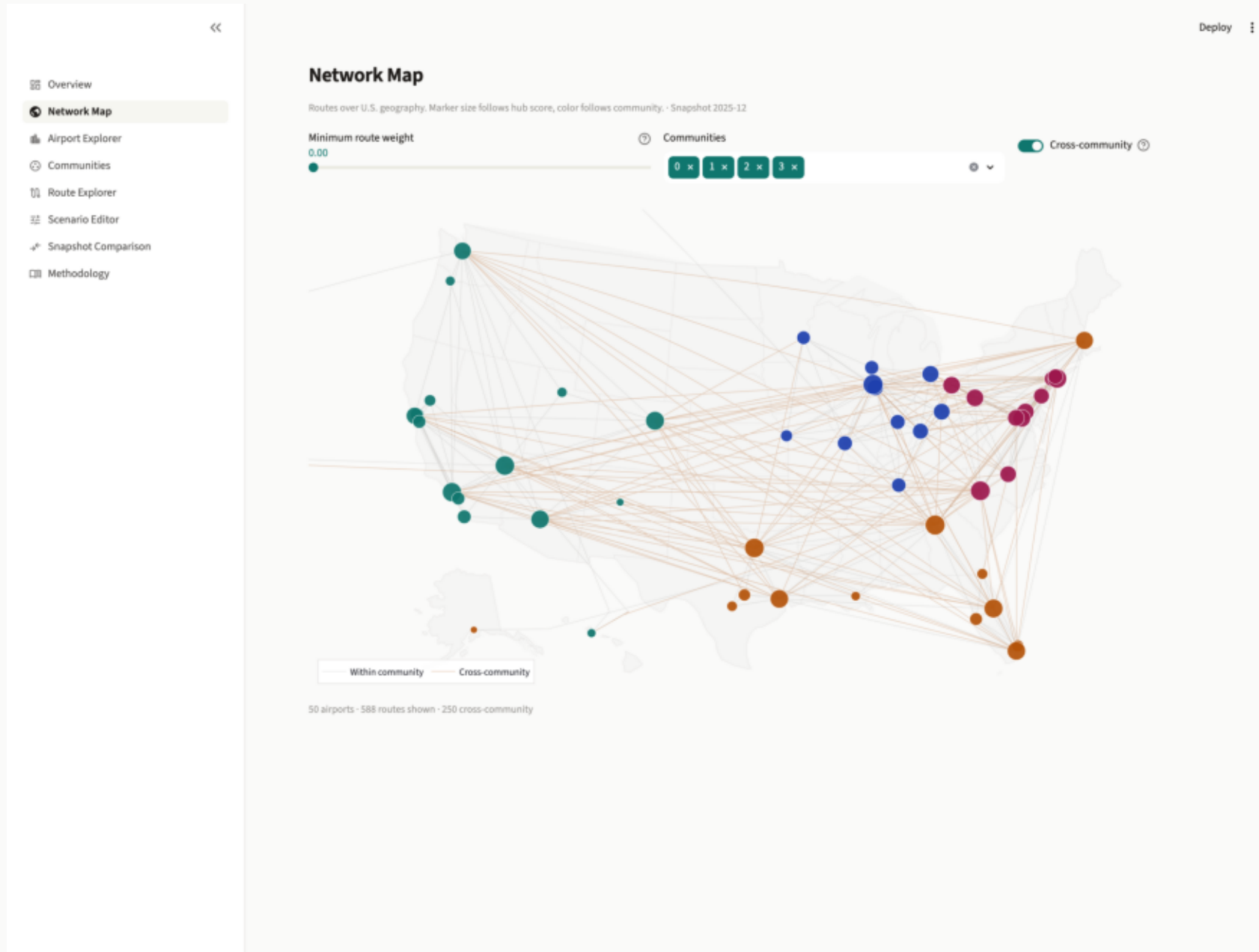
Overview

Snapshot totals, then the projects central claim: hub score against bridge score with the diagonal drawn in. Airports above the line carry more structural load than their traffic implies.



Network Map

The route network over an Albers USA projection. Marker area follows hub score, color follows Leiden community, and cross-community routes are tinted separately. The regional structure is legible directly from the map.



Airport Explorer

Every airport in the snapshot, searchable by code, name, or city. Percentile scores render as bars on a fixed 0-100 domain so standing is comparable between tables.

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Overview

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Airport Explorer

Communities

Route Explorer

Scenario Editor

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Airport Explorer

Every airport in the snapshot, ranked and filterable by structural role. - Snapshot 2025-12

Search

Code, name, or city

0 x

1 x

2 x

3 x

Communities

Sort by

Vulnerability

Desc

Code	Airport	City	State	Community	Hub	Bridge	Vulnerability	PageRank	Betweenness
ORD	Chicago O'Hare International	Chicago, IL	IL	2	99.4	100.0	100.0	0.0442	
ATL	Hartsfield-Jackson Atlanta International	Atlanta, GA	GA	1	99.0	98.0	98.0	0.0459	
CLT	Charlotte Douglas International	Charlotte, NC	NC	3	96.0	94.0	95.2	0.0399	
DFW	Dallas/Fort Worth International	Dallas/Fort Worth, TX	TX	1	94.0	96.0	94.8	0.0356	
LAX	Los Angeles International	Los Angeles, CA	CA	0	89.0	92.0	89.6	0.0315	
LAS	Harry Reid International	Las Vegas, NV	NV	0	92.4	86.0	88.4	0.0354	
JFK	John F Kennedy International	New York, NY	NY	3	89.4	80.0	87.2	0.0298	
MCO	Orlando International	Orlando, FL	FL	1	86.0	88.0	85.6	0.0295	
DEN	Denver International	Denver, CO	CO	0	84.4	82.0	84.4	0.0292	
PHX	Phoenix Sky Harbor International	Phoenix, AZ	AZ	0	79.6	84.0	82.8	0.0280	

50 airports

Airport detail

ORD · Chicago O'Hare International

Chicago O'Hare International

ORD · Chicago, IL, IL · community 2

Vulnerability

Bridge

Hub

100.0

100.0

99.4

DEPARTURES

17,509

TOTAL DEGREE

54

ARRIVALS

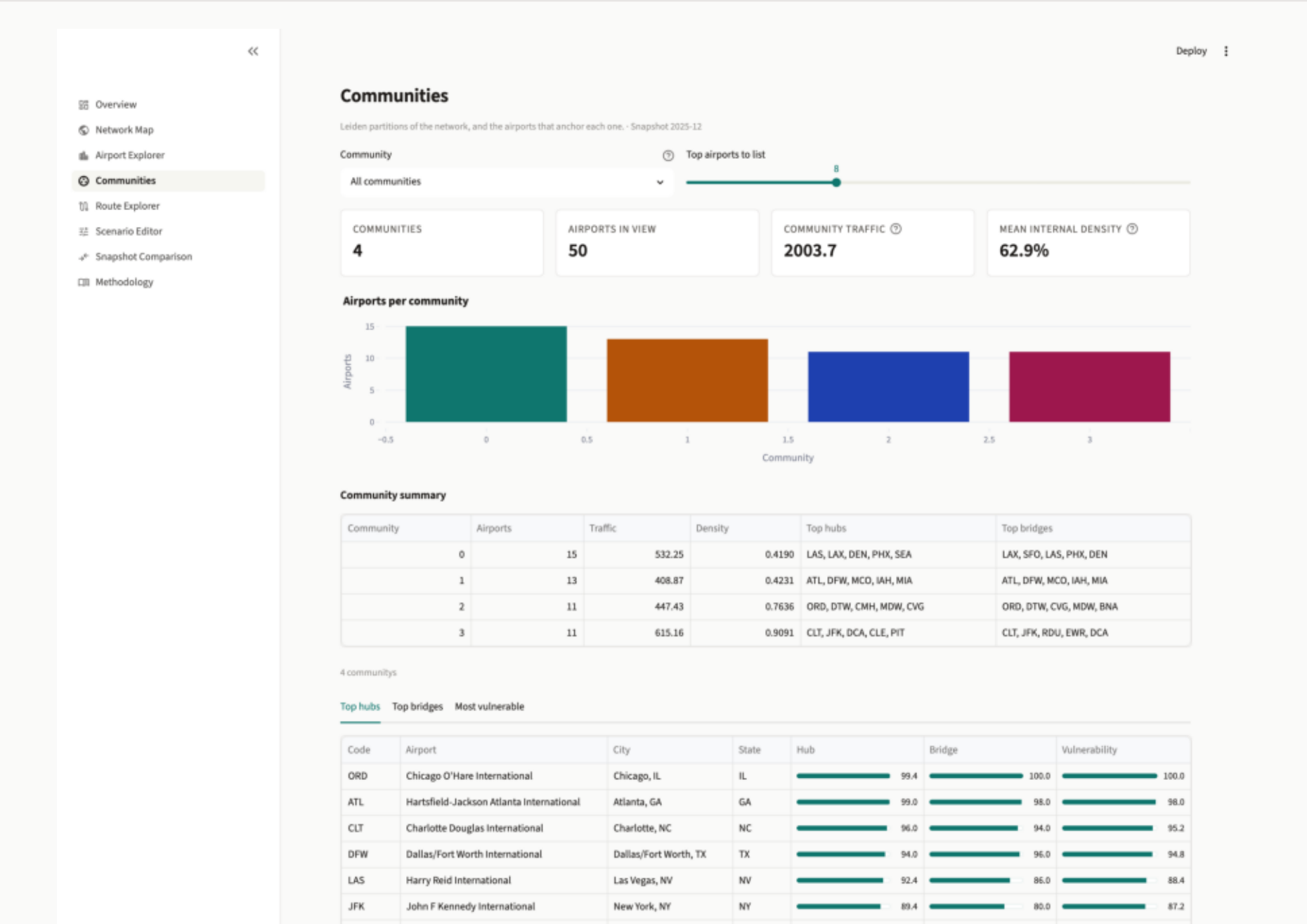
17,494

PAGERANK

0.04418

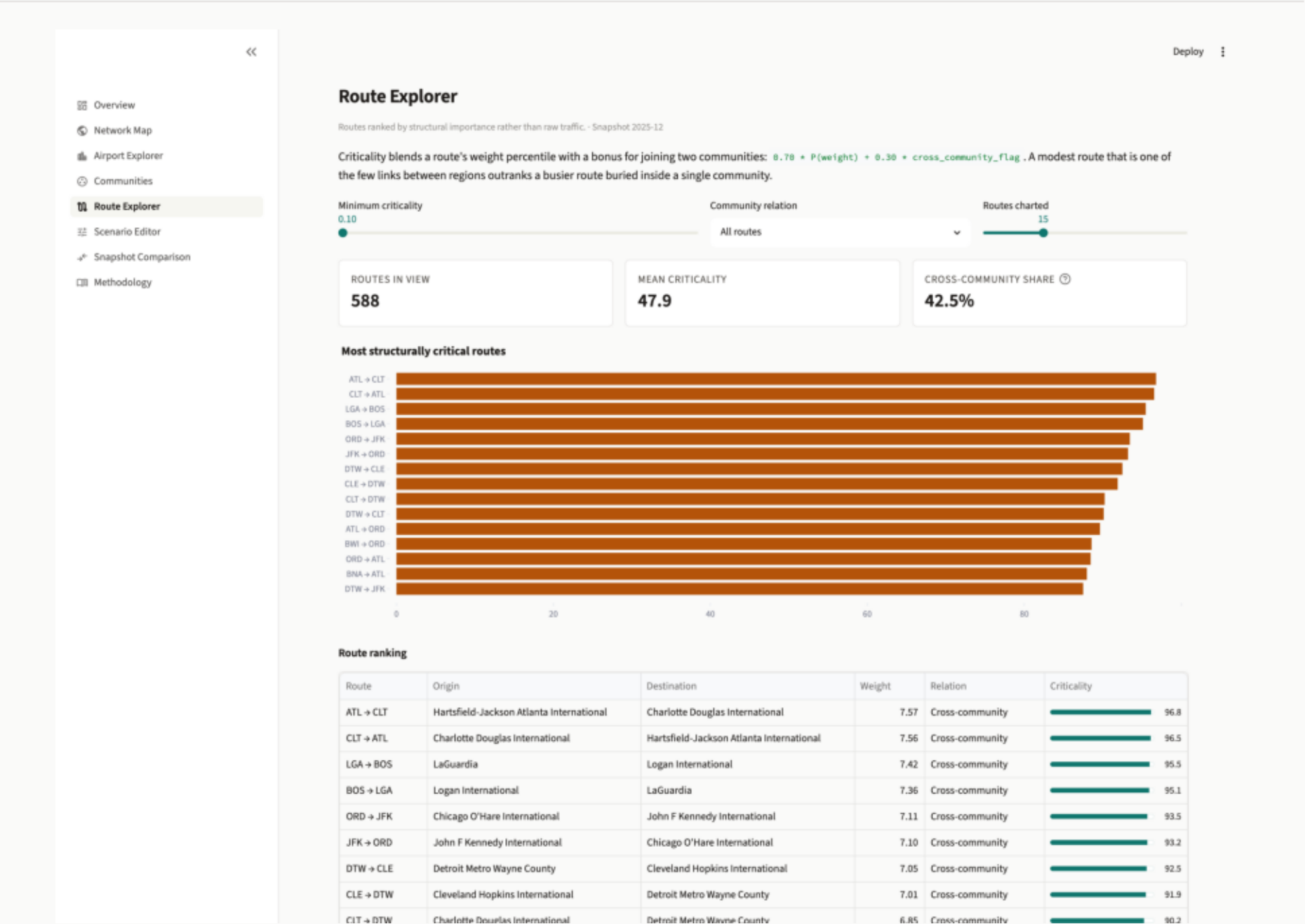
Communities

Leiden partitions with their size, internal traffic, and density. The top hub and bridge lists stored as airport ids in the artifact are resolved to codes for display.



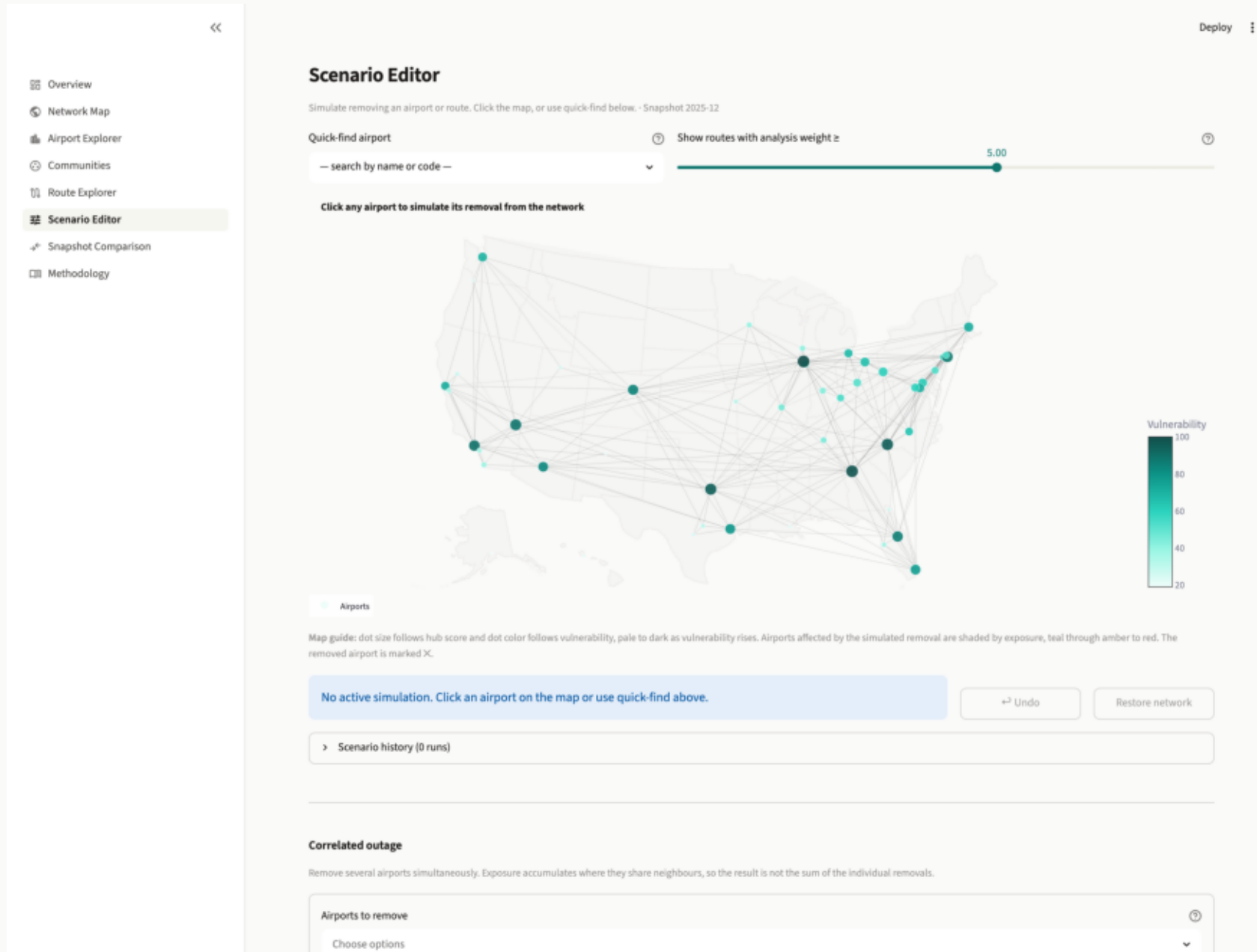
Route Explorer

Routes ranked by structural importance rather than raw traffic. A modest route that is one of few links between regions outranks a busier route buried inside one community.



Scenario Editor

The baseline network before any simulation. Any airport can be removed by clicking it, or through the quick-find control.



Snapshot Comparison

Whether the scores describe the network or one month's sampling. Rank agreement between any two built snapshots, and a slope chart of the airports that moved most, since the aggregate correlation hides exactly those cases.

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Snapshot Comparison

Whether the scores describe the network or just one month of sampling.

Baseline
2022-11

Compare against
2022-12

AIRPORTS IN BOTH
345

ONLY IN 2022-11
3

ONLY IN 2022-12
1

Rank agreement

Spearman correlation between the two months. A structural metric should reproduce its own ranking; a score that does not is describing the sample.

Score	Spearman rho	Verdict
Hub score	0.97	stable
Bridge score	0.91	stable
Vulnerability	0.86	mostly stable

What moved

The exceptions are the useful part, because the aggregate correlation hides them. A seasonal airport *should* move; a mover with no plausible explanation is worth investigating.

Airports to show

8

Largest hub-score movements

Airport	Nov 2022	Dec 2022
EGG	35	65
LTN	35	62
STN	35	60
BHX	35	38
LHR	35	35
MAL	35	32
GVA	35	30
AMS	35	28
CDG	35	25
FRA	35	22
DCA	35	20
JFK	35	18
LAX	35	15
HND	35	12
PVG	35	10

Methodology

Every formula as implemented, followed by the model's known limitations stated plainly rather than omitted.

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Methodology

Data/model assumptions, formulas, and caveats used by this implementation.

Data pipeline and graph model

- Snapshot policy:** Monthly snapshots (`snapshot_id = YYYY-MM`) from processed artifacts.
- Primary graph:** Directed weighted airport network where edge weight is `analysis_weight`.
- Weight definition:** `analysis_weight = log1p(flight_count)`.
- App behavior:** Pages read precomputed artifacts (`metrics.csv`, `communities.csv`, `route_metrics.csv`, `scenarios.csv`, `scenario_exposure.csv`) through app loaders with required-column guards.

Airport-level metrics

The metrics engine computes centrality and percentile-scaled composites for comparison:

$$P(\textit{metric}) = \text{percentile rank scaled to } [0, 100]$$
$$\textit{HubScore}(i) = 0.50 \cdot P(s_{\textit{total}}(i)) + 0.30 \cdot P(\textit{PageRank}(i)) + 0.20 \cdot P(\textit{deg}_{\textit{total}}(i))$$
$$\textit{BridgeScore}(i) = P(\textit{Betweenness}(i))$$
$$\textit{Vulnerability}(i) = 0.60 \cdot P(\textit{ImpactScore}(\textit{remove } i)) + 0.40 \cdot P(\textit{BridgeScore}(i))$$

Community and route metrics

Leiden communities partition airports into groups used by APP-04 and APP-05.

$$\textit{CommunityTraffic}(C) = \sum_{(i,j) \in C} w(i,j)$$
$$\textit{InternalDensity}(C) = \frac{\text{internal edges}(C)}{|C| \cdot (|C| - 1)}$$
$$\textit{RouteCriticality}(i,j) = 0.70 \cdot P(w(i,j)) + 0.30 \cdot \textit{CrossCommunity}(i,j)$$

`CrossCommunity(i,j)` is implemented as:

- `100` when origin and destination belong to different Leiden communities

Scenario Editor — active simulation

ATL removed. The removed airport is marked, airports touched by the two-hop ripple are shaded by exposure, and the impact and network-health scores update against the live graph.

